CAMBRIDGE A2 LEVEL MATHEMATICS (9709) • UNIT 0.01

C A2-Level 9709 Paper 3 & Paper 6 Diagnostic Readiness

PRACTICE WORKSHEET

Pure 3 & Probability/Statistics 2 Readiness • Cambridge 9709

Name: _____ **Date:** _____ **Score:** _____ / 25

A2-Level 9709 Paper 3 & Paper 6 Diagnostic Readiness Practice Exercises

1 MCQ

Which of the following is the correct expression of $\frac{5x+1}{(x-1)(x+2)}$ in partial fractions?

A $\frac{2}{x-1}+\frac{3}{x+2}$

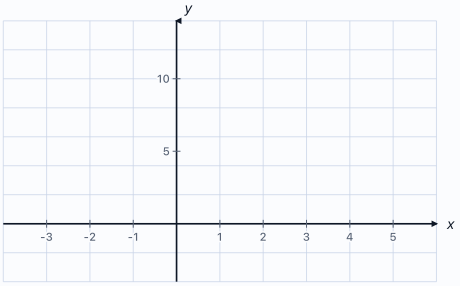
B $\frac{3}{x-1}+\frac{2}{x+2}$

C $\frac{2}{x-1}-\frac{3}{x+2}$

D $\frac{1}{x-1}+\frac{4}{x+2}$

2 MCQ

A curve has equation $y=\ln(2x^3+1)$. What is the gradient of the tangent to the curve at the point where $x=1$?



A 2

B 6

C $\frac{2}{3}$

D $\ln 3$

3 MCQ

Find the exact value of the definite integral

$$\int_0^{\ln 2} e^{2x}\,dx.$$

A $\frac{3}{2}$

B 3

C $\frac{1}{2}$

D $\frac{7}{2}$

4 MCQ

The rational function f is defined for all real $x \notin \{-1, 2\}$ by

$$f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$$

where k is a constant. The partial fraction decomposition of $f(x)$ takes the form

$$A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$$

where A, B, C, and D are real constants.
Given that $C = 2$, find the value of k.

A -3

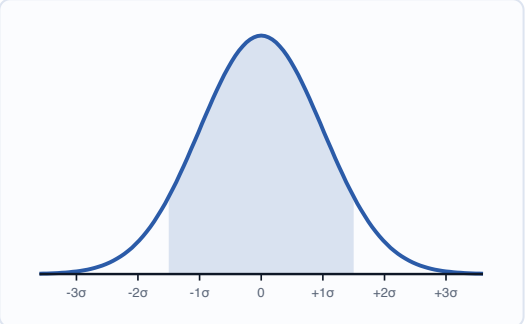
B 1

C 3

D 24

5 MCQ

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$.
The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$.
Using the standard normal critical value $z_{0.05} = 1.645$:
1. Find the value of σ.
2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.



A 43

B 11

C 22

D 44

6 MCQ

The complex number z satisfies the equation

$$z(2-\mathrm{i}) = 7+4\mathrm{i}.$$

Express z in the form $x+\mathrm{i}y$, where x and y are real.

A $2+3\mathrm{i}$

B $\frac{18}{5}+\frac{1}{5}\mathrm{i}$

C $2-3\mathrm{i}$

D $\frac{10}{3}+5\mathrm{i}$

7 MCQ

A continuous random variable X has probability density function given by $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, and $f(x) = 0$ otherwise. Find $\mathrm{P}(1 \le X \le 2)$.

A $\frac{7}{8}$

B $\frac{1}{8}$

C $\frac{3}{4}$

D $\frac{1}{2}$

8 MCQ

A hypothesis test is carried out at the 10% significance level. Find the probability of a Type I error.

A 90\%

B 50\%

C 10\%

D 0\%

9 MCQ

A hypothesis test is carried out at the 2% significance level. Find the probability of a Type I error.

A 0\%

B 98\%

C 2\%

D 50\%

10 MCQ

In an applied problem, a hypothesis test is carried out at the 2.5% significance level. Find the probability of a Type I error.

A 2.5\%

B 50\%

C 0\%

D 97.5\%

11 MCQ

Suppose a hypothesis test is carried out at the 1% significance level. Find the probability of a Type I error.

A 50\%

B 99\%

C 0\%

D 1\%

12 MCQ

An observed hypothesis test is carried out at the 5% significance level. Find the probability of a Type I error.

A 50\%

B 5\%

C 95\%

D 0\%

13 MCQ

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses

$$H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$$

at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

A $z = -2.0$; reject H_0 because $-2.0 < -1.645$

B $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

C $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

D $z = -8.0$; reject H_0 because $-8.0 < -1.645$

14 MCQ

The continuous random variable X has probability density function $f(x)$ defined by

$$f(x) = \begin{cases} kx & 0 \le x < 2, \\ k(6 - 2x) & 2 \le x \le 3, \\ 0 & \text{otherwise}, \end{cases}$$

where k is a positive constant. Find the exact value of the conditional probability $\mathrm{P}(X > 1 \mid X < 2.5)$.

A $\dfrac{9}{11}$

B $\dfrac{10}{11}$

C $\dfrac{3}{4}$

D $\dfrac{9}{10}$

15 MCQ

Calls arrive at a technical support desk according to a Poisson process with an average rate of λ calls per hour. It is known that the probability of receiving exactly 1 call in a given hour is 1.5 times the probability of receiving exactly 3 calls in that same hour.
Find the probability that the desk receives exactly 4 calls during a randomly chosen 3-hour period.

A $\frac{2187}{8}\mathrm{e}^{-9} \approx 0.0337$

B $54\mathrm{e}^{-6} \approx 0.134$

C $\frac{625}{24}\mathrm{e}^{-5} \approx 0.175$

D $\frac{2}{3}\mathrm{e}^{-2} \approx 0.0902$

16 MCQ

A wire of length 4 m is bent to form a rectangle with adjacent sides of lengths X m and $(2-X)$ m, where X is a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{3}{8}x^2, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Given that the area of the rectangle is at least $\frac{3}{4}$ m^2, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.

A $\frac{37}{351}$

B $\frac{127}{640}$

C $\frac{74}{351}$

D $\frac{127}{351}$

17 MCQ

A quality control protocol uses a two-stage hypothesis test to monitor the rate of flaws, λ per metre, in the production of high-grade optical fibre. The number of flaws in a length of fibre follows a Poisson distribution. The manufacturer tests:

$$H_0 : \lambda = 1 \quad \text{against} \quad H_1 : \lambda > 1.$$

The two-stage procedure operates as follows:
- **Stage 1:** A 1-metre sample is inspected, and the number of flaws X_1 is recorded. If $X_1 \ge 3$, H_0 is rejected immediately. If $X_1 = 0$, H_0 is accepted immediately. If $X_1 \in \{1, 2\}$, the test proceeds to Stage 2.
- **Stage 2:** A second, independent 1-metre sample is inspected, and the number of flaws X_2 is recorded. If $X_1 + X_2 \ge 4$, H_0 is rejected; otherwise, H_0 is accepted.
Given that the true flaw rate is $\lambda = 2$ per metre, find the probability that a Type II error is committed.

A $16\mathrm{e}^{-4} \approx 0.293$

B $4\mathrm{e}^{-2} \approx 0.541$

C $1 - \mathrm{e}^{-1} - \frac{7}{2}\mathrm{e}^{-2} \approx 0.158$

D $\mathrm{e}^{-2} + 16\mathrm{e}^{-4} \approx 0.428$

18 MCQ

Flaws in an optical fiber occur according to a Poisson process with a constant mean rate of λ flaws per kilometer. An automated quality-assurance test inspects two independent sections of fiber. The first section has length $1\ \text{km}$ and the second section has length $2\ \text{km}$. Let X and Y denote the number of flaws observed in the first and second sections, respectively. Given that the total number of flaws observed across both sections is $X+Y=3$, find the conditional probability $\mathrm{P}(X \geq 1 \mid X+Y=3)$.

A $\frac{8}{27}$

B $\frac{7}{8}$

C $\frac{19}{27}$

D $\frac{2}{3}$

19 MCQ

Two independent random samples, each of size n, are taken from a normal population with mean μ and known standard deviation σ. From each sample, a symmetric 90% confidence interval for μ is constructed. Given that $\Phi(1.645)=0.9500$ and $\Phi(2.326)=0.9900$, where Φ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

A 0.980

B 0.900

C 0.810

D 0.990

20 MCQ

The continuous random variable X has probability density function given by

$$f(x)=\begin{cases} k(4-x^2) & 0 \leq x \leq 2, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.
Find the value of k.

A $\frac{3}{16}$

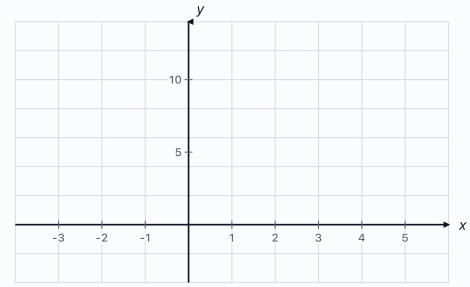
B $\frac{3}{8}$

C $\frac{16}{3}$

D $\frac{1}{4}$

21 MCQ

Referring to the coordinate curve shown in the diagram, a hypothesis test at the 1% significance level gives a p-value of 6%. What is the decision?



A Do not reject H_0

B Test is inconclusive

C Reject H_0

D Increase the sample size

22 MCQ

For a hypothesis test at the 5% (one-tail) significance level, find the critical z-value.

A 1.96

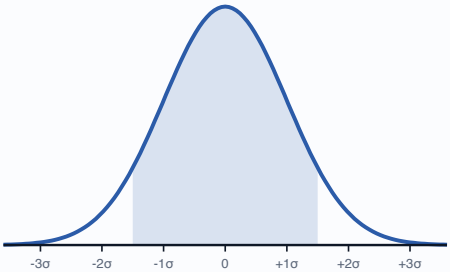
B 1.282

C 2.576

D 1.645

23 MCQ

For a hypothesis test at the 5% (two-tail) significance level, find the critical z-value.



A 2.576

B 1.645

C 1.282

D 1.96

24 MCQ

A hypothesis test at the 1% significance level gives a p-value of 4%. What is the decision?

A Reject H_0

B Increase the sample size

C Test is inconclusive

D Do not reject H_0

25 MCQ

A hypothesis test at the 10% significance level gives a p-value of 15%. What is the decision?

A Increase the sample size

B Do not reject H_0

C Reject H_0

D Test is inconclusive